MedVisionQuery: An Agentic Multimodal System for Medical Visual Question Answering with Explanation

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Abstract

Medical visual question answering (VQA) over radiology images combines perception, question interpretation, answer synthesis and verification—steps that fail in different ways and are hidden inside a single model call. We present MedVisionQuery, an agentic system in which a Haystack Agent controller decides, case by case, how to look at the image, answer, verify the answer *against the image*, and whether to re-examine a specific region, revise, abstain or finish. Perception is question-aware, the answerer and the verifier both see the image, and a different, higher-tier model verifies. We evaluate on 300 questions sampled at random from the whole VQA-RAD dataset against a single-prompt baseline, a hand-designed workflow and a fixed-pipeline ablation, all on the same answering model, scoring with a strict rule-based scorer and a blinded LLM judge. The single prompt is more accurate than the agentic loop (64.0% vs. 59.7% strict, 70.0% vs. 64.7% judged; McNemar $p = 0.079$ and $p = 0.033$, significant only under the judge), the loop and the fixed pipeline do not differ significantly ($p = 0.58$), and the loop's higher-tier verifier answering alone beats it (67.0%, $p = 0.008$). On the questions it answers, the loop matches the single prompt (65% vs. 66%); its 25 abstentions account for the gap. Its value is calibrated confidence: answers its check verifies are 71% correct, the rest 28%—at about 58× the cost of a single prompt. Code is available at `https://github.com/clarence-ck/med-vqa-rad-agentic`.

1 Problem Definition and Motivation

Medical VQA is a fundamentally multi-stage reasoning problem [1]. As established in the literature, a complete medical VQA system must address at minimum an image encoder, a question encoder, a multimodal fusion component, and an answer prediction module [1]—but these four components conceal distinct reasoning sub-problems that interact in clinically important ways. Given a radiology image and a clinical question, a system must separately solve: (1) *perception*—what is visible in the image that bears on the question; (2) *question interpretation*—what type of reasoning and answer is required, since different question types demand fundamentally different skills [2, 3]; (3) *synthesis*—how to integrate visual evidence into a grounded answer; (4) *verification*—whether the answer is actually supported by the visual evidence, a stage explicitly modelled in recent work as a "Confirm" module for cross-modal validation [5]; and (5) *control*—deciding, from what verification finds, whether to re-examine the image, revise the answer, abstain or stop.

These sub-tasks interact in non-trivial ways, which motivates making them explicit steps: what to look at depends on the question; the verifier must check cross-modal consistency between the image and the generated answer; and clinical safety requires explicit guardrails, abstention and disclaimers when confidence is low.

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Existing work: classification approaches. Early and still-prevalent medical VQA methods frame the task as multi-class classification over a fixed answer vocabulary [1]. The VQA-Med challenge [6] established the dominant paradigm: an image encoder and a question encoder produce fused features passed to a classification head. MEVF [7] extended this with meta-learning (MAML) and a denoising autoencoder (CDAE), feeding image features into a bilinear attention network (BAN) followed by a classifier. PMC-VQA [8] proposed MedVInT-TE, similarly mapping image–question pairs to answers via an encoder and classification head. Classification approaches are computationally efficient on close-ended questions but constrained to answers seen during training and perform poorly on open-ended questions requiring free-form responses.

Existing work: generation approaches. To handle open-ended questions, a parallel line of work adopts autoregressive generation. Bazi et al. [9] proposed a full transformer encoder–decoder architecture demonstrating gains on VQA-RAD and PathVQA. PMC-VQA's MedVInT-TD [8] further demonstrated that an LLM decoder can match or exceed classification-based performance when fine-tuned on large biomedical corpora. LLaVA-Med [10], accepted at NeurIPS 2023, adapted the general-domain LLaVA architecture to biomedicine via curriculum learning on PMC-15M figure–caption pairs and GPT-4-generated instruction-following data. RadFM [11] scaled this into a generalist radiology foundation model capable of VQA, report generation, and rationale diagnosis. R-LLaVA [12] extended LLaVA-Med by injecting clinician-annotated regions of interest into the CLIP encoder, improving spatial and contextual comprehension.

Existing work: hybrid approaches. Q2ATransformer [2] proposed a unified transformer framework introducing learnable candidate answer embeddings as a decoder, querying the fused image–question representation for each answer class. Although framed as classification, candidate answer embeddings interact through transformer attention, incorporating aspects of generative reasoning. This hybrid achieves state-of-the-art performance on VQA-RAD and PathVQA.

The gap that remains. These approaches—whether classification [6–8], generation [9–12], or hybrid [2]—answer in a single end-to-end model invocation, which makes it hard to localise failures, adapt the reasoning to what the image shows, or enforce clinical safety constraints. Recent work adds explicit structure: MedVCTP [5] is a training-free See–Think–Confirm pipeline whose last stage checks cross-modal consistency, and Qiu et al. [4] ground answers in a chain of evidence; MedVCTP's stages, however, run in a fixed order. MedVisionQuery instead uses an agent controller that coordinates question-aware perception, answering and image-grounded verification, deciding per case whether to re-examine the image, revise, abstain or finish, with structured failure handling.

We target the **VQA-RAD** benchmark [13], a clinically generated radiology VQA dataset; its Hugging Face release contains 2,244 image–question pairs, each with a reference answer, across close-ended and open-ended question types. Because no system is trained, we sample the evaluation questions from the whole dataset rather than only its test split (Section 5).

2 Agent Architecture

Controller loop. A fresh Haystack `Agent` is created per request (Figure 1). Its five tools are bound to per-request working memory holding the image, the question analysis, and every look, answer and check; the controller sees only short text results and chooses the next call. Unlike the fixed-pipeline ablation, the tool sequence depends on what verification finds: on the 300 evaluation cases the controller finished directly in 187 cases, re-examined a region in 63, revised its answer in 25 and abstained in 25. Budgets cap the loop at two focused re-looks, three answers and 16 steps, with a safety net that answers or checks before finishing if the controller stalls.

Roles. The *controller* is the planner: it chooses every next step from the results so far. The per-request *working memory* (image, question analysis, looks, answers, checks) is the memory; the tools read from and write to it. `analyze_question` (task understanding) classifies the question type and expected answer format. `look(focus)` (perception) examines the image *with the question in mind*; with a focus, it zooms: the named region is located, cropped and enlarged, and examined beside the full image; the first look rejects non-radiology images. `answer(guidance)` (executor) answers from the image plus all observations, applying the verifier's guidance on revisions. `check` (evaluator, a higher-tier model) has two parts. A *blind reading*, made once per episode: the verifier answers

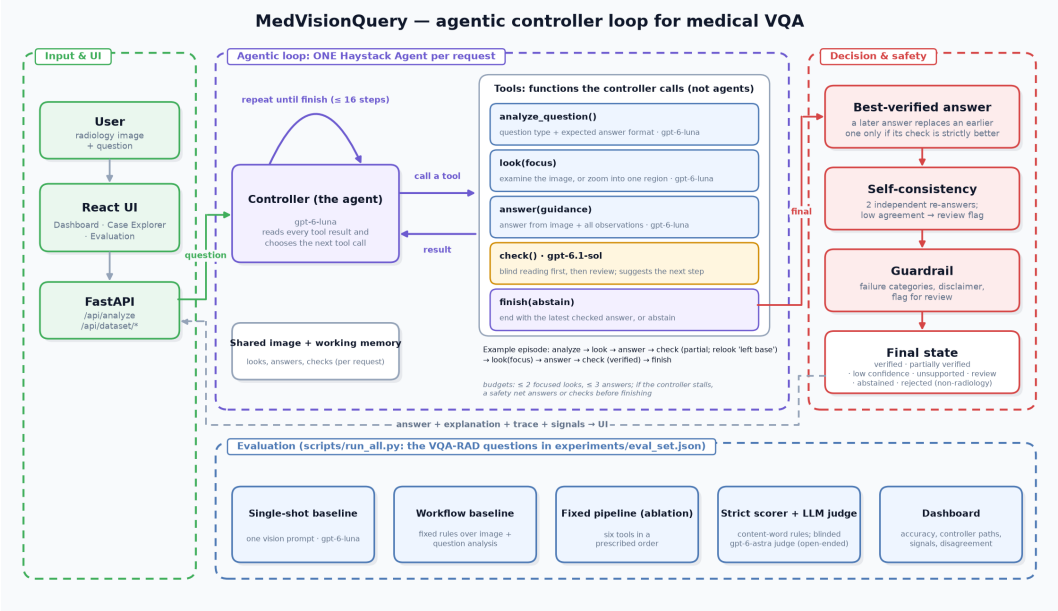


Figure 1: MedVisionQuery. One Haystack Agent (the controller) calls five tools in a loop, reading each result before choosing the next: question analysis, question-aware perception (`look`), answer synthesis, image-grounded verification (`check`) and `finish`. The tools are not agents; `check` only recommends a re-look, a revised answer, abstention or acceptance, and the controller decides. A decision-and-safety stage keeps the best-verified answer, measures self-consistency and applies guardrails.

from the image and question alone, without the proposed answer or the looker's notes. A *review*: it checks the answer against the image (grounding, contradictions, format) and returns a status (verified, partially verified, low confidence, unsupported), the issues found and a next step—*accept*, *relook* with a focus, *reanswer* with guidance, or *abstain*; the guidance may include a suggested corrected answer. An answer is verified only if the review passes and the blind reading agrees (by word rules, or for open-ended answers by meaning, judged by the verifier); the controller learns only whether they agree. `finish(abstain)` ends the episode; a normal finish is refused while the latest answer is unchecked.

Model and tool choices. All roles use OpenAI models, sized by need and configurable per role. The controller and question analysis make simple decisions (`gpt-6-luna`, no/low reasoning effort); perception and answering use `gpt-6-luna` with medium effort, and the answerer always sees the image, because a text description loses the visual detail many questions depend on. Verification uses a different, higher-tier model (`gpt-6.1-sol`, 20× the per-token price) so that the answering model never verifies itself; since its guidance can suggest answers, we also score it answering alone as a reference. Haystack's `Agent/Tool` abstraction provides the tool-calling loop and exit conditions; keeping the image in per-request memory rather than in the controller's context keeps the controller cheap.

Baselines and ablation. The *single-shot baseline* sends the image and question in one prompt. The *workflow baseline* is a fixed chain—a question-agnostic image description, question analysis, then hand-written rules per question type. The *fixed-pipeline ablation* is a controller instructed to call six tools in a prescribed order—`interpret_image`, `analyze_question`, `filter_image_context`, `route_difficulty`, `reason_answer`, `verify_answer`—re-running the sequence once when verification flags a fixable problem. It uses the same models but a question-agnostic image description, no zoom, no blind check and a text-only verifier, so comparing against it measures the agentic design as a whole, not control alone. A demo UI (Section 6) exposes all systems.

Table 1: Accuracy (%) on 300 random VQA-RAD questions (same answering model for all systems). Open-ended accuracy is strict/judged. Time is per-case latency in seconds (mean/median), measured with four cases running concurrently; the whole run of all four systems took 67 minutes of wall-clock time. Cost is the mean API cost per case.

System	Strict	Judged	Close (160)	Open (140)	Time (s)	Cost
Single-shot baseline	64.0	70.0	81.9	43.6 / 56.4	4.3 / 3.6	$0.0002
Workflow baseline	28.0	29.3	44.4	9.3 / 12.1	9.3 / 8.8	$0.0005
Agentic loop	59.7	64.7	73.8	43.6 / 54.3	48.1 / 34.2	$0.0132
Fixed pipeline (ablation)	58.0	62.0	73.1	40.7 / 49.3	34.5 / 27.9	$0.0068

3 Multimodal Grounding

The system uses two modalities, the radiology image and the text question, and they interact at every step that determines the answer: the question (and any focus) steers what `look` examines; `answer` and `check` receive the image directly alongside the text observations, because a text description of the image loses the visual detail they depend on; only question analysis and the controller work from text. The evaluation's LLM judge compares answers with the reference as text. Because the verifier sees the image, its "explanation grounded in evidence" and "no contradictions" checks are genuine cross-modal agreement checks.

4 Control and Failure Handling

Detection. Image-grounded verification (five checks and a status), seven failure categories (*unsupported claim*, *low confidence*, *visual contradiction*, *partially verified*, *question–answer mismatch*, *generation unstable*, *parser failure*) and an eighth, *self-inconsistent*, from resampling. **Recovery.** Instead of re-running the whole pipeline, the controller zooms into the region the verifier names or revises the answer with the verifier's guidance. When the blind reading disagrees, the controller is told only *that* it disagrees, never the blind answer, so the loop must resolve it with its own model. A later answer replaces an earlier one only if its check is strictly better, because focused re-looks can over-read. **Self-critique.** After finishing, the answerer is re-asked independently twice; low agreement adds a failure category and flags the case for review. **Fallback.** The controller may abstain (reported as an abstention, never coerced into "no"); a guardrail adds a disclaimer and a review flag for low-confidence, unsupported, high-severity or self-inconsistent cases; non-radiology or unreadable images are rejected before any answer is produced.

5 Evaluation

Setup. 300 questions drawn at random from the whole VQA-RAD dataset, keeping its close/open-ended mix (seed 51511; 160 close-ended, 140 open-ended; one known-bad item and an earlier evaluation's 300 questions excluded). 277 carry the train label, so scores are not directly comparable with published test-split results. All four systems run on the same cases with the same answering model. *Strict scoring*: yes/no answers must start with the correct word; binary terms must name the correct term and not its opposite; counts must match; open-ended answers must contain every content word of the reference (aliases and plurals folded; $\geq$60% for references longer than three words), with left/right conflicts and denials counted wrong and embedding similarity [14] only as a strict fallback. Because word matching both rejects correct paraphrases and accepts vague answers that contain the reference word, a *blinded LLM judge* (`gpt-6-astra`, a model that neither answers nor verifies in any system; answers shuffled and anonymised) scores every open-ended answer as a second, independent measure; a 30-verdict spot check agreed with 29 credit decisions (the other, strict, on an agentic answer). Paired comparisons use the exact McNemar test; 95% intervals are about $\pm$5.6 points. The baseline and agentic loop were re-run on 100 cases to measure stability. API cost of the evaluation: $8.73 ($6.20 runs, $1.31 repeats, $1.22 judge). Wall-clock time: 67 min for the main run (four cases at a time), 83 min for the sequential repeats and 11 min for the judge.

Results. Table 1. The single prompt is ahead of the agentic loop (30 vs. 17 discordant cases, $p = 0.079$; judged 33 vs. 17, $p = 0.033$); the loop and the fixed pipeline do not differ significantly

Table 2: Left: accuracy (strict, %) by the path the controller took. Right: agentic accuracy by its own signals.

Path	n	Agentic	Single	Pipeline	Signal	Agentic
direct	187	75	74	70	verified (220)	71
relook	63	38	41	33	partially verified (65)	34
reanswer	25	56	64	52	unsupported/low (15)	0
abstained	25	–	44	40	consistency high (242)	69
					consistency med/low (33)	39–40

($p = 0.58$; judged $p = 0.35$); all three far exceed the workflow. The verifier model answering alone—the check's blind reading—scores 67.0% strict and beats the loop ($p = 0.008$). The gap to the single prompt comes from abstention: on the 275 questions it answered, the loop scores 65% versus 66%, while on its 25 abstentions the single prompt was right in 11 and the loop's own last answer in 7. The loop's mechanisms work individually: revision fixed 12 wrong first answers and broke 5 (88 revised episodes), and when the blind check objected to a first answer it was wrong in 61 of 82 cases. They do not lift the hardest cases above the single prompt: zoomed re-looks score 33% vs. 38% and re-answered cases 56% vs. 64% (Table 2). A post-hoc check (rule chosen after seeing these results, not a system result) locates the loss: taking the blind reading whenever it disagreed with the loop's final answer would have raised the loop's word-rule score by about 7 points, to the verifier's own.

Failure case analysis. *Self-consistency:* two independent re-answers per case; high consistency means 69% accuracy, medium/low 39–40%, so low consistency marks uncertainty rather than certain error. *Cross-modal agreement:* because the verifier sees the image, verification status is predictive: verified answers are 71% correct, the rest 28%. *Verifier model:* a higher-tier model verifies the answering model; after its checks, 113 of 300 episodes leave the direct path. *Disagreement:* the three systems disagree on 69% of cases; when exactly one system is right, it is the agentic loop 15 times, the single prompt 20 and the workflow 12. *Stability:* on 100 repeated cases, 6% of single-prompt answers flipped between right and wrong, versus 12% for the agentic loop. Five patterns dominate the errors: (i) a shared perception blind spot: 65 questions (22%), mostly open-ended ones asking for a specific location, finding or diagnosis, defeat the single prompt, the loop and the pipeline alike; no check catches a misreading that looker and verifier share, and 51 of the 65 still get high self-consistency; (ii) abstention withholding a right answer: the verifier's review recommended 21 of the 25 abstentions, 14 times although its own blind reading agreed with the loop's answer; (iii) the hardest questions resisting re-looks, with an occasional revision breaking a right answer; (iv) benchmark quirks—ungrammatical questions, or references needing detail an image rarely settles—where abstaining or hedging is scored wrong; and (v) workflow rigidity, where fixed rules answer "Abdomen" or "Chest" when a specific finding or side is needed.

6 User Interface

A React frontend exposes three views: **Dashboard** (upload an image, compare the agentic loop with both baselines, inspect the trace, verification checks and failure categories), **Case Explorer** (browse the 300 evaluation cases with full agent traces) and **Evaluation** (strict and judged accuracy for all four systems, accuracy by question type and by controller path, disagreement and a per-case table).

7 Reflection

When agentic behaviour is beneficial. On this benchmark, not for accuracy: the single prompt is more accurate (significantly under the judge) and the loop trails its own verifier answering alone. Its checking finds errors—revision fixes more answers than it breaks, and the blind check's objections are usually right—but its decisions waste that signal: it abstains, or keeps its own answer, where the verifier's blind reading is often right, and a loop repaired by deferring to that reading would be little more than its verifier. Its answers are also less stable between runs. Its benefit is *calibrated confidence*: the verified label separates answers that are 71% correct from the rest at 28%, and every decision is auditable, which suits review workflows that can act on that signal. Answering

with a low-confidence flag instead of abstaining, and letting a disagreeing blind reading replace a low-confidence answer, tested on fresh questions, are the natural next steps.

Limitations. About 58× the cost and 10× the median latency of a single prompt, for lower accuracy on this benchmark. Abstentions are scored as wrong, so the abstention policy lowers accuracy by design. The verifier shares the perception weaknesses of vision models, so agreement is not correctness, and because its guidance may suggest answers, the loop's contribution cannot be fully separated from the verifier's (a critique-only verifier would isolate it). 300 questions give ±5.6-point intervals and noisy per-type estimates; the LLM judge is itself a model, spot-checked but not human-annotated. The system uses no retrieval, so it cannot draw on knowledge that is not visible in the image.

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